

AWS Storage Services

Comprehensive Learning Guide

AWS Backup · Amazon EBS · Amazon EFS · Amazon S3

Beginner-Friendly · In-Depth · Practical · Hands-On



AWS Backup

Centralised Backups



Amazon EBS

Block Storage



Amazon EFS

File Storage



Amazon S3

Object Storage

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■ AWS Backup

Centralised, Fully Managed Backup Service Across All AWS Resources

1. Simple Introduction

AWS Backup is a **fully managed, centralised backup service** that makes it easy to automate and centrally manage backups of data across AWS services — including EC2, EBS, RDS, DynamoDB, EFS, FSx, S3, and even on-premises resources via Storage Gateway. Without AWS Backup, teams write custom scripts or manually manage snapshots per service — fragile, inconsistent, and hard to audit.

■ **Real-World Analogy:** AWS Backup is like a **bank's centralised vault management system**. Instead of each branch managing its own safe independently with different schedules and lock types, the central vault system enforces the same backup policy, audit log, and retention rule across every branch automatically — from one control panel.

Business Problems Solved:

- Eliminate manual, per-service backup scripting — one policy governs everything
- Meet compliance mandates (GDPR, HIPAA, PCI-DSS) with centralised audit logs and retention
- Recover from ransomware or accidental deletion quickly with vaulted, immutable backups
- Enforce backup policies across all accounts in an AWS Organisation from one place
- Protect on-premises workloads (VMware, Storage Gateway) alongside cloud resources

2. Core Concepts

Term	Definition	Example
Backup Plan	A policy defining when to back up, how long to retain, and which vault to use	Daily backup at 3 AM, retain 35 days, move to cold after 90
Backup Rule	A component of a backup plan defining schedule, retention, and lifecycle	Cron(0 3 * * ? *), retain 35 days
Backup Vault	An encrypted container that stores and organises recovery points	production-vault, compliance-vault
Recovery Point	A backup of a resource at a specific point in time (snapshot/copy)	RDS snapshot of mydb taken 2025-06-15 03:00 AM
Backup Selection	Which resources are backed up by a plan (by tag, ARN, or resource type)	All resources with tag: backup=true
Backup Window	Time period within which the backup job must start	Start within 1 hour of 3 AM
Lifecycle Policy	Automatically transition recovery points to cold (cheap) storage after N days	Move to cold after 30 days; expire after 365 days
Vault Lock	Write-once-read-many (WORM) lock on a vault — immutable backups	Compliance-vault locked; no one can delete backups for 7 years

[illegible]

Step	What Happens
1. Plan Creation	Admin creates Backup Plan with rules (schedule, retention, vault, lifecycle)
2. Resource Selection	Assign resources to plan by tag (e.g. backup=true), ARN, or resource type
3. Job Trigger	Scheduler triggers backup jobs per the cron/rate schedule
4. Snapshot/Copy	AWS Backup calls native snapshot APIs for each service (e.g. RDS CreateSnapshot)
5. Store in Vault	Recovery point stored encrypted in the specified vault
6. Lifecycle	After N days, recovery point automatically moved to cold storage or expired
7. Cross-Copy	Optional: copy recovery point to another region/account for DR
8. Restore	On demand: restore any recovery point to a new resource in minutes

4. Features

Feature	Description
Centralised Backup Console	Single pane of glass for backup jobs, plans, vaults, recovery points across all services
Tag-Based Backup Selection	Apply a tag (backup=true) and any matching resource is automatically protected
Backup Vault Lock (WORM)	Compliance mode: no one — not even root — can delete backups during lock period
Cross-Region Backup Copy	Automatically copy recovery points to a DR region for geo-redundancy
Cross-Account Backup Copy	Isolate backup storage in a separate AWS account — protects from account-level incidents
Lifecycle Management	Automatically tier recovery points from warm to cold (S3 Glacier) to save costs
Backup Audit Manager	Generate compliance reports (e.g. backup frequency report) for auditors

Feature	Description
Organisation-Level Policies	Enforce backup policies across all AWS accounts in an Organisation — non-overridable
Restore Testing	Automatically restore backups and validate them — test without human intervention
VMware Workload Support	Back up on-premises VMware VMs to AWS via Backup Gateway
Continuous Backup (PITR)	Point-in-time recovery for RDS, Aurora, DynamoDB, S3 — restore to any second
EventBridge Integration	Backup job state changes trigger EventBridge events for automation/alerting

5. Real-World Use Cases

Industry	Use Case
Banking	Centralised daily backup of 500 RDS instances, 200 EFS filesystems, and 1,000 EBS volumes across 20 accounts — one policy
Healthcare	HIPAA: Vault Lock ensures patient record backups are immutable for 7 years; Audit Manager generates evidence for auditors
E-Commerce	Cross-region backup of DynamoDB order tables and RDS product catalogue — restore < 1 hour if primary region fails
Enterprise	Organisation-level backup policy ensures every EC2 and RDS in all 50 AWS accounts has a daily backup — no exceptions
Startup	Tag-based backup: add backup=true tag to any new resource and it's automatically protected — no DevOps needed
Government	Compliance-mode vault lock on all audit data for 10-year retention — meets FedRAMP requirements
Retail	Restore Testing: nightly validation that yesterday's RDS backup can be restored successfully — proves recoverability
Manufacturing	VMware on-premises ERP backed up to AWS via Backup Gateway — single console for on-prem and cloud

6. Practical Example

Scenario: Create a backup plan that backs up all resources tagged backup=true daily, retains for 35 days, copies to a DR region, and locks the vault.

```
aws backup create-backup-vault \ --backup-vault-name prod-vault \ --encryption-key-arn
arn:aws:kms:us-east-1:123:key/mrk-abc aws backup put-backup-vault-lock-configuration \
--backup-vault-name prod-vault \ --min-retention-days 7 \ --max-retention-days 365 \
--changeable-for-days 3 aws backup create-backup-plan --backup-plan '{ "BackupPlanName":
"prod-daily-plan", "Rules": [{ "RuleName": "daily-3am", "TargetBackupVaultName": "prod-vault",
"ScheduleExpression": "cron(0 3 * * ? *)", "StartWindowMinutes": 60, "CompletionWindowMinutes":
180, "Lifecycle": { "MoveToColdStorageAfterDays": 30, "DeleteAfterDays": 365}, "CopyActions": [{
"DestinationBackupVaultArn": "arn:aws:backup:eu-west-1:123:backup-vault:dr-vault", "Lifecycle":
{"DeleteAfterDays": 90} }] } ]' PLAN_ID=$(aws backup list-backup-plans --query
```

```
'BackupPlansList[0].BackupPlanId' --output text) aws backup create-backup-selection \
--backup-plan-id $PLAN_ID \ --backup-selection '{ "SelectionName": "tag-backup-true",
"IamRoleArn": "arn:aws:iam::123:role/AWSBackupRole", "ListOfTags":
[{"ConditionType": "STRINGEQUALS", "ConditionKey": "backup", "ConditionValue": "true"}] }'
```

■ **Result:** Any resource tagged backup=true is backed up daily at 3 AM, retained 35 days warm + 335 days cold, copied to eu-west-1, and protected by Vault Lock — fully automated with zero per-service scripts.

7. Integrations

AWS Service	Integration
Amazon EC2 / EBS	Backs up EC2 instances (AMI) and EBS volumes (snapshots) on schedule
Amazon RDS / Aurora	Creates automated RDS snapshots and supports point-in-time recovery (PITR)
Amazon DynamoDB	On-demand backups and continuous PITR (restore to any second in last 35 days)
Amazon EFS	Backs up EFS file systems; restores to same or new EFS filesystem
Amazon S3	Backs up S3 buckets with PITR; restore individual objects or entire buckets
Amazon FSx	Backs up FSx for Windows, Lustre, ONTAP, and OpenZFS filesystems
AWS KMS	All recovery points encrypted with KMS — customer-managed or AWS-managed keys
Amazon EventBridge	Backup job lifecycle events (started, completed, failed) trigger automation
Amazon SNS	Notify ops teams on backup job failure via SNS topics
AWS CloudTrail	Audit log of all AWS Backup API calls for compliance evidence
AWS Organizations	Enforce backup policies across all accounts from the management account
AWS Storage Gateway	Back up on-premises Storage Gateway volumes to AWS Backup

8. Comparison

Feature	AWS Backup	Native Snapshots (per-service)	AWS DLM	Third-Party (Veeam)
Scope	All AWS services in one place	One service at a time	EBS volumes only	Multi-cloud / on-prem
Cross-account	■ Yes	■ No (manual copy)	■ No	■ Yes
Cross-region	■ Yes (auto copy rules)	Manual copy per snapshot	■ Yes (EBS only)	■ Yes
Vault Lock	■ WORM compliance mode	■ Not available	■ No	■ Vendor-specific
Audit reports	■ Audit Manager built-in	■ No	■ No	■ Yes

Feature	AWS Backup	Native Snapshots (per-service)	AWS DLM	Third-Party (Veeam)
Org policies	■ Central across all accounts	■ Per account	■ Per account	Agent-based
PITR	■ RDS, DDB, S3, Aurora	Varies by service	■ Snapshot only	Varies
On-premises	■ VMware via Gateway	■ No	■ No	■ Yes
Cost	Per GB-month of backup storage	Per GB-month of snapshots	Free (pay for EBS snaps)	License + storage
Best for	Centralised enterprise backup policy	Simple per-resource snapshots	EBS lifecycle automation	Hybrid/multi-cloud

■ Amazon Elastic Block Store (EBS)

High-Performance, Persistent Block Storage for Amazon EC2

1. Simple Introduction

Amazon EBS (Elastic Block Store) is a **high-performance, persistent block storage service** designed for use with Amazon EC2 instances. EBS volumes behave exactly like raw, unformatted hard drives — you can format them with a filesystem, install an OS, run databases, or store any data. The key difference from an internal hard drive is that EBS volumes persist independently of the EC2 instance — they survive instance stops, reboots, and even termination (if configured).

■ **Real-World Analogy:** EBS is like an **external USB hard drive for your EC2 server**. You can plug it in (attach), use it as fast local storage, unplug it (detach), and plug it into a different server (reattach to another EC2). Unlike a physical USB drive, EBS is replicated within an Availability Zone for durability, and you can take point-in-time snapshots of it at any time.

Business Problems Solved:

- Persistent OS and data volumes that survive EC2 instance reboots and stops
- High-IOPS, low-latency block storage for databases (MySQL, PostgreSQL, Oracle, Cassandra)
- Consistent performance with provisioned IOPS — predictable sub-millisecond latency
- Point-in-time snapshots for backup, DR, and volume cloning
- Encrypt sensitive data at rest with AWS KMS with zero performance impact

2. Core Concepts

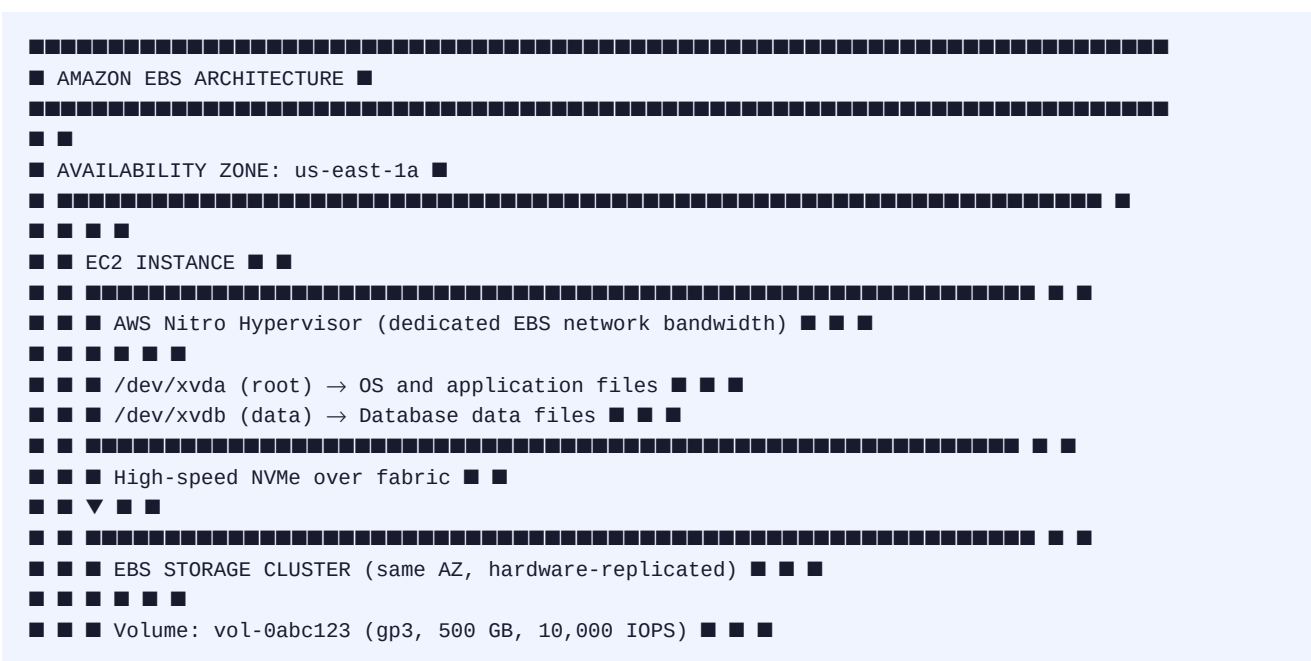
Term	Definition	Example
Volume	A block storage device that is created and attached to an EC2 instance	100 GB gp3 volume attached to a web server
Volume Type	The performance and cost characteristics of the volume	gp3 (general), io2 (high IOPS), st1 (throughput), sc1 (cold)
IOPS	Input/Output Operations Per Second — measure of storage performance	3,000 IOPS (gp3 default) vs 64,000 IOPS (io2 max)
Throughput	MB/s of sustained data transfer the volume can sustain	125 MB/s (gp3 default) vs 1,000 MB/s (gp3 max)
Snapshot	A point-in-time incremental backup of an EBS volume stored in S3	Snapshot of 100 GB volume taken at midnight
Multi-Attach	Attach a single io1/io2 volume to multiple EC2 instances simultaneously	Shared storage for clustered applications (Oracle RAC)
Root Volume	The EBS volume containing the OS — /dev/xvda or /dev/sda1	30 GB gp3 root volume with Amazon Linux 2023
Data Volume	Additional EBS volumes attached for application data	500 GB io2 volume for PostgreSQL data files

Term	Definition	Example
Encryption	AES-256 encryption of data at rest, in transit, and in snapshots	KMS-encrypted volume — zero performance overhead
Availability Zone	EBS volumes exist in a single AZ — must be in same AZ as EC2	us-east-1a volume attached to us-east-1a EC2 only
Elastic Volumes	Modify volume type, size, and IOPS without detaching or stopping	Grow 100 GB to 500 GB while app is running
Fast Snapshot Restore	Pre-initialise snapshots for instant full performance on restore	No I/O slowdown when restoring a new volume from snapshot

EBS Volume Types Comparison

Volume Type	Full Name	Max IOPS	Max Throughput	Use Case	Pricing Model
gp3	General Purpose SSD v3	16,000	1,000 MB/s	Boot, dev, most workloads	Per GB + IOPS/throughput separately
gp2	General Purpose SSD v2	16,000	250 MB/s	Legacy general-purpose	Per GB (IOPS based on size)
io2 Block Express	Provisioned IOPS SSD	256,000	4,000 MB/s	Mission-critical DBs, SAP HANA	Per GB + per IOPS
io1	Provisioned IOPS SSD v1	64,000	1,000 MB/s	High-perf databases	Per GB + per IOPS
st1	Throughput Optimised HDD	500	500 MB/s	Big data, Kafka, log processing	Per GB (cheap)
sc1	Cold HDD	250	250 MB/s	Archival, infrequent access	Lowest per GB

3. Architecture & Working



[illegible]

Lifecycle Step	Action
1. Create Volume	Specify type, size, IOPS, throughput, AZ, encryption key — ready in seconds
2. Attach	aws ec2 attach-volume — volume appears as a block device in OS
3. Format & Mount	mkfs.xfs /dev/xvdb && mount — now usable as a filesystem
4. Use	Applications read/write — EBS replicates within AZ transparently
5. Snapshot	Incremental snapshot to S3 — first is full, subsequent are delta-only
6. Detach / Reattach	Detach from one EC2, reattach to another in same AZ
7. Modify (Elastic)	Change size, type, or IOPS without downtime — no restart needed
8. Delete	Permanent deletion — unless set to retain on EC2 termination

4. Features

Feature	Description
Elastic Volumes	Dynamically increase size, change type (gp2→gp3), adjust IOPS while in use
Snapshots	Incremental point-in-time backups stored durably in S3 — use for backup, DR, cloning
Snapshot Lifecycle (DLM)	Amazon Data Lifecycle Manager automates snapshot creation and deletion schedules
Fast Snapshot Restore	Pre-warm snapshot so restored volumes deliver full performance immediately
Encryption	256-bit AES encryption for all data at rest and in transit — managed by KMS
Snapshot Copy	Copy snapshots across regions for DR; share snapshots with other AWS accounts
Multi-Attach (io1/io2)	Attach one volume to up to 16 Nitro EC2 instances in same AZ concurrently

Feature	Description
EBS-Optimised Instances	Dedicated network bandwidth between EC2 and EBS — eliminates network contention
Nitro SSD	io2 Block Express backed by AWS-designed custom NVMe silicon — 256K IOPS
CloudWatch Metrics	VolumeReadOps, VolumeWriteOps, BurstBalance, VolumeQueueLength for monitoring
Recycle Bin	Protect accidentally deleted snapshots — recover within retention period
Archive Tier	Move cold snapshots to EBS Snapshot Archive — 75% cheaper, 24-72 hr restore

5. Real-World Use Cases

Industry / Workload	EBS Volume Choice	Details
Web Server OS Boot	gp3 (20-30 GB)	Fast boot, low cost; OS and app binaries on root volume
MySQL / PostgreSQL	gp3 or io2 (100 GB+)	Database data, redo logs, temp files — needs consistent low latency
Oracle RAC	io2 Multi-Attach	Shared block storage across multiple EC2 Oracle nodes
SAP HANA (in-memory DB)	io2 Block Express	256K IOPS, sub-ms latency — only block storage certified for SAP HANA on AWS
Kafka Brokers	st1 (1-16 TB)	Sequential writes at high throughput — HDD is sufficient and much cheaper
ML Training Data	gp3 (1-16 TB)	Staging area for training datasets before feeding into GPU instances
Video Editing Workstation	io2 (2 TB)	Low-latency random access to large video files
CI/CD Build Cache	gp3 (100 GB)	Fast access to build artefacts and dependency caches

6. Practical Example

Scenario: Create a high-performance database volume, attach it to EC2, format it, and set up automated daily snapshots.

```
aws ec2 create-volume \ --volume-type gp3 \ --size 500 \ --iops 10000 \ --throughput 500 \
--availability-zone us-east-1a \ --encrypted \ --kms-key-id arn:aws:kms:us-east-1:123:key/mrk-db \
--tag-specifications 'ResourceType=volume,Tags=[{Key=Name,Value=db-data-vol}]' aws ec2
attach-volume \ --volume-id vol-0abc1234567890def \ --instance-id i-0123456789abcdef0 \ --device
/dev/xvdb # On the EC2 instance (SSH in): sudo mkfs.xfs /dev/xvdb sudo mkdir -p /data/postgres
sudo mount /dev/xvdb /data/postgres echo '/dev/xvdb /data/postgres xfs defaults,nofail 0 2' | sudo
tee -a /etc/fstab aws dlm create-lifecycle-policy \ --description 'Daily DB snapshots 35-day
retention' \ --state ENABLED \ --execution-role-arn arn:aws:iam::123:role/DLMRole \
--policy-details '{ "PolicyType":"EBS_SNAPSHOT_MANAGEMENT", "ResourceTypes":["VOLUME"],
"TargetTags":[{"Key":"Name","Value":"db-data-vol"}], "Schedules":[{"
"Name":"daily","CreateRule":{"CronExpression":"cron(0 3 * * ? *)"}, "RetainRule":{"Count":35},
"CopyTags":true } ] }'
```

■ **Result:** 500 GB gp3 volume with 10,000 IOPS and 500 MB/s throughput attached and formatted. Daily snapshots at 3 AM with 35-day retention — fully automated.

7. Integrations

AWS Service	Integration
Amazon EC2	EBS is the primary persistent storage for EC2 — root and data volumes
AWS Backup	Central backup service can back up EBS volumes on schedule
Amazon Data Lifecycle Manager	Automate EBS snapshot creation, retention, and cross-region copy
AWS KMS	Encrypt EBS volumes and snapshots with customer-managed KMS keys
Amazon CloudWatch	Monitor volume IOPS, throughput, latency, queue depth metrics
AWS CloudTrail	Audit all EBS API calls (CreateVolume, AttachVolume, DeleteSnapshot)
Amazon ECS / EKS	EBS volumes used as persistent storage for containerised workloads
Amazon RDS	RDS uses EBS under the hood for database storage (io1/gp3 types)
AWS Elastic Disaster Recovery	EBS replication between regions for application DR
AWS Systems Manager	Automate snapshot creation via SSM Run Command or Maintenance Windows

8. Comparison

Feature	EBS	Instance Store	Amazon EFS	Amazon S3
Storage Type	Block	Block (ephemeral)	File (NFS)	Object
Persistence	■ Persists	■ Lost on stop	■ Persists	■ Persists
Scope	Single AZ	Single EC2	Multi-AZ	Global (regional)
Access	Single EC2 (or Multi-Attach io2)	Single EC2	Multiple EC2 simultaneously	HTTP/SDK/CLI
Performance	Up to 256K IOPS	Up to millions IOPS	General file workloads	Throughput-optimised
Resize	■ Live resize	Fixed at launch	■ Elastic auto	Unlimited (auto)
Snapshots	■ Yes	■ No	■ Via AWS Backup	■ Versioning
Encryption	■ KMS	■ Hardware	■ KMS	■ SSE-S3/KMS/C
Best for	DBs, OS volumes	Temp HPC scratch	Shared file system	Object/data lake

■ Amazon Elastic File System (EFS)

Scalable, Shared, Serverless NFS File Storage for AWS Compute

1. Simple Introduction

Amazon EFS (Elastic File System) is a **fully managed, serverless, elastic NFS (Network File System) file storage service** that automatically grows and shrinks as you add and remove files — no provisioning required. Multiple Amazon EC2 instances, ECS containers, Lambda functions, or on-premises servers can access the same EFS filesystem simultaneously, making it ideal for shared workloads.

■ **Real-World Analogy:** EFS is like a **shared network drive in an office**. Every employee's computer (EC2 instance, container, Lambda) can connect to the shared drive simultaneously. If the drive gets full, it automatically expands — you never have to buy bigger hardware. Files saved by one user are instantly visible to all others. And unlike a physical server room, this drive is replicated across multiple buildings (AZs) automatically.

Business Problems Solved:

- Share the same files across hundreds of EC2 instances without complex synchronisation
- No capacity planning — EFS scales from kilobytes to petabytes automatically
- Serverless file storage — no filesystem administration, patching, or resizing
- Lift-and-shift NFS-dependent applications without changing application code
- Provide a shared home directory or content repository for all compute instances

2. Core Concepts

Term	Definition	Example
File System	The top-level EFS resource — a managed NFS namespace	fs-0abc123456789def0 in us-east-1
Mount Target	An NFS endpoint in a VPC subnet used by compute to connect to EFS	10.0.1.45 in us-east-1a, 10.0.2.72 in us-east-1b
Access Point	Application-specific entry point with enforced directory, UID/GID, and POSIX permissions	Access point /app-data enforces UID 1000 for all connections
Performance Mode	General Purpose (low latency) or Max I/O (high throughput, higher latency)	General Purpose for CMS; Max I/O for big data analytics
Throughput Mode	Bursting (tied to size), Provisioned (fixed MB/s), or Elastic (auto-scales)	Elastic for unpredictable workloads; Provisioned for consistent needs
Storage Class	Standard (multi-AZ, high durability) or One Zone (single AZ, cheaper)	Standard for production; EFS One Zone for dev/test
Lifecycle Policy	Automatically move files not accessed in N days to EFS Infrequent Access (IA) storage	Move files not accessed in 30 days to IA — 92% cheaper
EFS Infrequent Access	Lower-cost storage tier for files not accessed frequently	IA cost: \$0.025/GB-month vs Standard \$0.30/GB-month

- Access Point /ml-data → enforces UID 2000, GID 2000 ■
- ■
- LIFECYCLE: ■
- Standard → (30 days no access) → IA → (90 days no access) → Archive ■
- ■
- ON-PREMISES ACCESS via AWS Direct Connect or VPN + mount target ■

Step	Action
1. Create Filesystem	Choose performance mode, throughput mode, lifecycle policy, encryption — done in seconds
2. Create Mount Targets	One mount target per AZ subnet — EFS assigns an IP in that subnet
3. Configure Security Group	Allow NFS port 2049 inbound from EC2 security groups
4. Mount on EC2	<code>sudo mount -t efs fs-0abc123:/mnt/efs</code> (or via <code>/etc/fstab</code> for persistence)
5. Use	Applications read/write normally — EFS handles concurrency and scale automatically
6. Lifecycle	EFS monitors last-access time; moves cold files to IA/Archive automatically
7. Backup	AWS Backup creates daily EFS backups; EFS Replication provides async DR copy

4. Features

Feature	Description
Fully Elastic Capacity	Grows and shrinks automatically from 0 to petabytes — no provisioning ever
Concurrent Access	Thousands of EC2, ECS, Lambda, and on-premises clients mount simultaneously
Elastic Throughput Mode	Automatically scales throughput up/down based on actual workload — recommended for variable workloads
Multi-AZ Durability	Standard class: data replicated across all AZs in the region — 99.999999999% durability
Intelligent Tiering	Lifecycle policies auto-move files to IA/Archive classes — reduce costs by up to 95%
EFS Replication	Automatic async replication to another region/AZ for DR with RPO in minutes
Access Points	Enforce POSIX user/group ID and root directory per application — multi-tenant isolation
IAM Authorization	Control NFS access with IAM policies and resource-based policies — no network-only security
EFS One Zone	Single-AZ EFS at ~47% lower cost — for dev/test or non-critical workloads

Feature	Description
On-Premises Access	Mount EFS from on-premises servers via Direct Connect or VPN
AWS DataSync Integration	Migrate data into EFS or sync between EFS filesystems
Container Native Support	EFS CSI driver for Kubernetes (EKS); ECS task definition volume mounts

5. Real-World Use Cases

Industry / Use Case	Details
Web Content Management	WordPress or Drupal running on 20 EC2 instances all sharing the same /var/www — uploads visible everywhere
Machine Learning Training	Training dataset stored on EFS; 50 p3.2xlarge EC2 instances mount simultaneously and read in parallel
Container Shared Storage	ECS tasks and EKS pods mount EFS for persistent shared state — e.g., a shared configuration or model store
Big Data / Analytics	Spark workers on EMR mount EFS as shared scratch space — Max I/O mode for high-throughput parallel reads
DevOps Build Cache	Shared build artefact cache on EFS — all Jenkins workers access the same cached dependencies
Home Directories	Centralised /home NFS mount for Linux developer workstations and bastion hosts
Video Rendering Farm	Render farm of 200 EC2 GPU instances all read raw footage and write rendered frames to shared EFS
Genomics Pipeline	Reference genome (hundreds of GB) stored on EFS; thousands of parallel analysis jobs read it concurrently
DR / Backup	EFS Replication to eu-west-1 as async DR target — failover by remounting from replica region

6. Practical Example

Scenario: Create an EFS filesystem with lifecycle tiering, mount it on two EC2 instances, and verify shared access.

```
aws efs create-file-system \ --performance-mode generalPurpose \ --throughput-mode elastic \
--encrypted \ --kms-key-id arn:aws:kms:us-east-1:123:key/mrk-efs \ --tags
Key=Name,Value=shared-app-fs FS_ID=fs-0abc123456789def0 aws efs put-lifecycle-configuration \
--file-system-id $FS_ID \ --lifecycle-policies \ '["TransitionToIA":"AFTER_30_DAYS"],{"Transition
ToArchive":"AFTER_90_DAYS"},{"TransitionToPrimaryStorageClass":"AFTER_1_ACCESS"]]' aws efs
create-mount-target \ --file-system-id $FS_ID \ --subnet-id subnet-aaa111 \ --security-groups
sg-efs-clients aws efs create-mount-target \ --file-system-id $FS_ID \ --subnet-id subnet-bbb222 \
--security-groups sg-efs-clients # On EC2 instance 1 (SSH): sudo mkdir -p /mnt/shared sudo mount
-t efs -o tls,iam $FS_ID:/ /mnt/shared echo 'Hello from EC2-1' | sudo tee /mnt/shared/test.txt #
On EC2 instance 2 (SSH) – same file visible immediately: cat /mnt/shared/test.txt # Output: Hello
from EC2-1 # Persistent mount via /etc/fstab: echo '$FS_ID:/ /mnt/shared efs _netdev,tls,iam 0 0'
| sudo tee -a /etc/fstab
```

■ **Result:** Two EC2 instances share the same filesystem. Files written on one are immediately visible on the other. Cold files automatically move to IA (92% cheaper) after 30 days. No capacity management needed —

EFS handles it all.

7. Integrations

AWS Service	Integration
Amazon EC2	Mount EFS via NFS — primary compute client for EFS
Amazon ECS	EFS volumes defined in task definitions for persistent shared container storage
Amazon EKS	EFS CSI driver enables persistent volume claims backed by EFS in Kubernetes
AWS Lambda	Lambda functions mount EFS for shared state and large dependency packages
AWS Batch	Batch jobs mount EFS to share large datasets across thousands of parallel jobs
AWS DataSync	Migrate data into EFS from on-premises NFS or S3; sync between EFS filesystems
AWS Backup	Back up EFS filesystems on a schedule; restore to same or new filesystem
AWS Transfer Family	SFTP/FTP/FTP server can use EFS as backend storage instead of S3
Amazon CloudWatch	Monitor EFS metrics: throughput, IOPS, connection count, storage bytes per tier
AWS KMS	Customer-managed encryption keys for EFS at-rest encryption
AWS Direct Connect	Mount EFS from on-premises via Direct Connect private virtual interface

8. Comparison

Feature	EFS	EBS	FSx for Windows	FSx for Lustre	Amazon S3
Protocol	NFS v4	Block (iSCSI-like)	SMB	Lustre (POSIX)	HTTP / S3 API
Access	Multi-instance	Single EC2*	Multi-instance	Multi-instance	Any client globally
Auto-scale	■ Automatic	Manual resize	Manual resize	Scratch: auto	■ Unlimited
Performance	General/Max I/O	Up to 256K IOPS	SSD 100K+ IOPS	Sub-ms, 100s GB/s	Throughput-optimised
Windows ACLs	■ POSIX only	Via OS	■ Full AD/NTFS	■ No	■ No
Lustre/HPC	■ No	■ No	■ No	■ Yes	■ No
Cost tier	IA/Archive tiers	One size	One size	Scratch/Persistent	■ Many tiers
Best for	Shared POSIX NFS	DB/OS block storage	Windows file shr	HPC workloads	Object data lake

■ Amazon Simple Storage Service (S3)

Infinitely Scalable Object Storage — The Foundation of AWS

1. Simple Introduction

Amazon S3 (Simple Storage Service) is AWS's **flagship object storage service** — infinitely scalable, highly durable, and globally accessible. You store files (called **objects**) in containers (called **buckets**). S3 stores any type and size of data — from a 1-byte text file to a 5 TB video — and serves it via simple HTTP requests. It underpins virtually every AWS service and is one of the most-used services in all of cloud computing.

■ **Real-World Analogy:** Amazon S3 is like a **global, infinitely large filing cabinet**. Each drawer is a bucket (labelled with a globally unique name). Inside, you file documents (objects) in any folder structure you like. The cabinet never runs out of space. Any authorised person in the world can retrieve a document via its address (URL). Certain drawers are public (anyone can read), others require a key (private/IAM-controlled).

Business Problems Solved:

- Store unlimited amounts of any data type without capacity planning or provisioning
- Serve static websites, images, videos, and downloads at global scale for pennies
- Data lake foundation — centralise all raw data for analytics and ML pipelines
- 99.999999999% (11 nines) durability — safest place to store any data
- Replace expensive on-premises file servers and tape archives with infinitely scalable cloud storage

2. Core Concepts

Term	Definition	Example
Bucket	A container for storing objects — globally unique name, region-specific	my-company-data-lake (must be globally unique across all AWS)
Object	A file and its metadata stored in S3 — up to 5 TB per object	images/product/sku-123.jpg (file + metadata + ETag)
Key	The unique identifier (path) of an object within a bucket	2025/06/15/orders.csv
Prefix	A simulated directory path — S3 is flat but keys create hierarchy	2025/06/15/ is a prefix containing multiple objects
Bucket Policy	Resource-based JSON policy controlling access to the bucket	Allow CloudFront to read all objects; deny public access
ACL	Access Control List — legacy per-object/bucket permissions (avoid; use policies)	Public-read ACL on a specific object
Versioning	Keep all versions of an object — enables recovery from accidental deletion	Delete v3 of orders.csv — v1 and v2 still exist
MFA Delete	Require MFA to permanently delete versioned objects or disable versioning	Root account + MFA device required to delete backup data

Term	Definition	Example
Storage Class	Cost/availability/latency tier for objects	Standard, IA, Glacier, Glacier Deep Archive, One Zone-IA
Lifecycle Policy	Rules to automatically transition or expire objects	After 30 days → IA; after 90 days → Glacier; after 365 → delete
Pre-Signed URL	Time-limited URL granting temporary access to a private object	URL valid for 1 hour; partner downloads invoice without AWS creds
S3 Transfer Acceleration	Routes uploads through CloudFront edge for faster global uploads	User in Tokyo uploads to US bucket 50% faster
Multipart Upload	Upload large objects in parallel parts — required > 5 GB, recommended > 100 MB	5 TB file uploaded as 10,000 × 500 MB parts in parallel
S3 Select	SQL query to retrieve subset of object data without downloading entire object	SELECT * FROM s3object WHERE price > 100 on a CSV file
Object Lambda	Transform data on-the-fly as it is retrieved from S3	Watermark images, redact PII, convert CSV to JSON on read
Event Notification	Trigger Lambda, SQS, SNS when objects are created, deleted, or restored	New file in uploads/ → triggers Lambda to process it
Replication (CRR/SRR)	Async replication to another bucket in same or different region	CRR: us-east-1 → eu-west-1 for compliance; SRR: prod → audit
Access Points	Named, per-application entry points with distinct access policies	analytics-ap with read-only access; ingest-ap with write-only
Object Lock	WORM protection on objects — Governance or Compliance mode	Financial records locked for 7 years — cannot be deleted

S3 Storage Classes

Class	Use Case	Availability	Min Duration	Retrieval	Cost
S3 Standard	Frequently accessed data	99.99%	None	Instant	Highest
S3 Standard-IA	Infrequent access, rapid restore	99.9%	30 days	Instant	Lower
S3 One Zone-IA	Re-creatable infrequent data	99.5%	30 days	Instant	Lower + 1 AZ risk
S3 Glacier Instant	Archive, millisecond restore	99.9%	90 days	Instant	Low
S3 Glacier Flexible	Archive, minutes-hours restore	99.99%	90 days	1 min-12 hr	Very low
S3 Glacier Deep Arc	Long-term archive 10+ years	99.99%	180 days	12-48 hrs	Lowest (\$0.00099/GB)
S3 Intelligent-Tier	Unknown/changing access pattern	99.9%	None	Instant	Monitoring fee + tier

Class	Use Case	Availability	Min Duration	Retrieval	Cost
S3 Express One Zone	Single-AZ, 10x faster latency	99.95%	None	1ms	Higher, lower request cost

Operation	HTTP Method	How It Works
HeadObject	HEAD	Retrieve metadata only (size, ETag, storage class) without body
CreateMultipart	POST	Initiate large object upload — upload parts in parallel, then complete

4. Features

Feature	Description
Unlimited Storage	No storage limit — store from 0 bytes to exabytes in a single bucket
11 Nines Durability	99.999999999% durability — lose 1 object in 10,000 years for 10M objects
Versioning	Keep all versions of every object — undelete or roll back at any time
Object Lock (WORM)	Prevent object deletion/overwrite for compliance — Governance or Compliance mode
Intelligent-Tiering	Automatically move objects between tiers based on access patterns — no retrieval fees
Lifecycle Policies	Automatically transition objects between storage classes or expire them
Cross-Region Replication	Async replicate objects to another region — compliance, DR, latency
S3 Access Logs	Log every request to your bucket for security and audit analysis
Static Website Hosting	Host static HTML/CSS/JS websites directly from S3 — no servers needed
Server-Side Encryption	SSE-S3 (free), SSE-KMS (auditable), SSE-C (customer-managed) — all objects encrypted
S3 Object Lambda	Transform objects on the fly as they're retrieved — mask PII, convert formats
S3 Batch Operations	Perform actions on billions of objects at once — copy, tag, restore, invoke Lambda
Block Public Access	Account-level and bucket-level setting to prevent accidental public exposure
Inventory Reports	Weekly/daily CSV or ORC/Parquet report of all objects and their metadata
Storage Lens	Organisation-wide analytics on S3 usage, activity, and cost optimisation
Mountpoint for S3	Open-source Linux client to mount S3 buckets as a local POSIX filesystem

5. Real-World Use Cases

Industry	Use Case	Storage Class
E-Commerce	Product images, videos, and CSS/JS assets served via CloudFront CDN	Standard

Industry	Use Case	Storage Class
Data Lake	Raw data from 50 microservices stored in S3 — Athena + Glue for querying	Standard + IA
ML / AI	Training datasets (images, text, audio) and model artefacts	Standard
Banking	7-year retention of financial transaction records for regulatory compliance	Glacier + Object Lock
Healthcare	DICOM medical images archived from PACS system — 99.999999999% durability	Glacier Instant
Media Streaming	Video segments served at scale via CloudFront — S3 as origin	Standard
Backup & DR	EBS snapshots, RDS exports, application backups centralised in S3	Standard-IA + Glacier
Gaming	Game patch distribution to millions of players worldwide via CloudFront	Standard
Log Archival	CloudTrail, VPC Flow Logs, ALB access logs — query with Athena	Standard-IA → Glacier
Static Website	React/Vue SPA hosted on S3 + CloudFront — zero server cost	Standard

6. Practical Example

Scenario: Create a secure data lake bucket with versioning, lifecycle tiering, replication, and event-driven processing.

```
aws s3api create-bucket \ --bucket my-company-data-lake \ --region us-east-1 \
--create-bucket-configuration LocationConstraint=us-east-1 aws s3api put-bucket-versioning \
--bucket my-company-data-lake \ --versioning-configuration Status=Enabled aws s3api
put-public-access-block \ --bucket my-company-data-lake \ --public-access-block-configuration \
'BlockPublicAcls=true,IgnorePublicAcls=true,BlockPublicPolicy=true,RestrictPublicBuckets=true' aws
s3api put-bucket-encryption \ --bucket my-company-data-lake \
--server-side-encryption-configuration '{ "Rules":[{"ApplyServerSideEncryptionByDefault":{"
"SSEAlgorithm":"aws:kms", "KMSMasterKeyID":"arn:aws:kms:us-east-1:123:key/mrk-s3"},
"BucketKeyEnabled":true}} ]}' aws s3api put-bucket-lifecycle-configuration \ --bucket
my-company-data-lake \ --lifecycle-configuration '{ "Rules":[{"
"ID":"tier-to-glacier","Status":"Enabled","Prefix":"raw/", "Transitions":[{"
"Days":30,"StorageClass":"STANDARD_IA"}, {"Days":90,"StorageClass":"GLACIER"},
{"Days":365,"StorageClass":"DEEP_ARCHIVE"} ] } ]}' aws s3api put-bucket-notification-configuration
\ --bucket my-company-data-lake \ --notification-configuration '{
"LambdaFunctionConfigurations":[{"
"LambdaFunctionArn":"arn:aws:lambda:us-east-1:123:function:process-upload",
"Events":["s3:ObjectCreated:*"],
"Filter":{"Key":{"FilterRules":[{"Name":"prefix","Value":"raw/"}]}} ]}'
```

■ **Result:** Secure, encrypted, versioned S3 data lake. Raw data auto-tiers: Standard → IA (30d) → Glacier (90d) → Deep Archive (365d). Every new object in raw/ triggers Lambda for processing. Block Public Access prevents accidental exposure.

7. Integrations

AWS Service	Integration
Amazon CloudFront	Use S3 as CDN origin — serve objects globally with low latency, OAC for private buckets
AWS Lambda	S3 event notifications trigger Lambda on object create/delete/restore
Amazon EventBridge	Advanced S3 event routing to any EventBridge target with filtering
Amazon Athena	Run SQL queries directly on S3 data (CSV, Parquet, JSON, ORC) — serverless analytics
AWS Glue	Crawl S3 data to build data catalogue; ETL jobs read/write S3
Amazon Redshift	COPY from S3 for bulk data load; UNLOAD to S3 for data exports
Amazon SageMaker	Training datasets and model artefacts stored in S3; SageMaker reads during training
AWS DataSync	Migrate on-premises files to S3; sync between S3 buckets
AWS KMS	SSE-KMS encrypts every object with customer-managed or AWS-managed key
Amazon CloudTrail	Log every S3 API call for security auditing — who accessed what, when
Amazon Macie	ML-based PII discovery and data security monitoring in S3 buckets
AWS Transfer Family	SFTP/FTPS/FTP server that stores files directly in S3 buckets
Amazon Rekognition	Analyse images/videos stored in S3 for object detection, face analysis
Amazon Transcribe/Comprehend	Process audio/text files from S3 for ML insights

8. Comparison

Feature	Amazon S3	Amazon EBS	Amazon EFS	Amazon Glacier	On-prem NAS
Type	Object storage	Block storage	File (NFS)	Archive object	File/Block
Access	HTTP/S3 API	EC2 block device	NFS mount	HTTP (delayed)	NFS/SMB
Concurrency	Unlimited clients	Single EC2*	Thousands	Read-only	Licensed
Max object size	5 TB	Volume size	File size	5 TB	Filesystem limit
Scaling	Unlimited	Up to 64 TB/vol	Petabytes	Unlimited	Hardware limit
Durability	99.999999999%	99.8–99.9%	99.999999999%	99.999999999%	Depends on RAID
Cold storage	■ Glacier tiers	Snapshot only	Archive tier	■ Native	Manual tiering
Static website	■ Native	■ No	■ No	■ No	■ No

Feature	Amazon S3	Amazon EBS	Amazon EFS	Amazon Glacier	On-prem NAS
Best for	Data lake, backup, web assets	DB/OS volumes	Shared NFS	Long-term archive	Legacy on-prem

Quick-Reference Summary

Service	Type	Key Strength	Best For
■ AWS Backup	Backup Management	Centralised policy + cross-account + Vault Lock	Enterprise backup governance across all services
■ Amazon EBS	Block Storage	High IOPS, low latency, provisioned performance	Databases, OS volumes, stateful EC2 workloads
■ Amazon EFS	File Storage (NFS)	Elastic shared NFS — thousands of clients at once	Shared filesystems, ML data, containers, content
■ Amazon S3	Object Storage	Unlimited scale, 11-nines durability, rich ecosystem	Data lake, backup, static assets, archival

Storage Decision Framework

```
START: What access pattern do you need?
█
██████████████████████████████████████████████████
█ █ █
▼ ▼ ▼
BLOCK STORAGE FILE STORAGE OBJECT STORAGE
(OS, databases, apps) (shared NFS/POSIX) (files via HTTP/API)
█ █ █
▼ ▼ ▼
Amazon EBS Amazon EFS Amazon S3
(single EC2*) (multi-EC2/ECS/ (unlimited, global,
gp3: general Lambda shared) HTTP access)
io2: high IOPS █
st1: throughput ████████████████████████████████████
█ █
Frequently Infrequently
accessed? accessed?
```

AND for ALL of the above:

- AWS BACKUP – centralise backup policy across ALL ■
- EBS, EFS, RDS, DynamoDB, S3, FSx from one console ■